

ARISTEIDIS TSAKNIS

Electrical & Computer Engineer | Junior Data Engineer

Thessaloniki, Greece
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Portfolio | LinkedIn | GitHub

PROFILE

Junior Data Engineer with hands-on experience building real-time data pipelines, ETL workflows, and automated ingestion systems using Python and SQL. Holds an Integrated Master's in Electrical & Computer Engineering and completed an internship developing efficient C++ software for decoding real-time IoT sensor data. Skilled in structured data processing, automation, and cloud-oriented architectures, with practical exposure to AWS through data engineering coursework. Additional background in applied machine learning and published research on autism detection using facial-image data.

EDUCATION

University of Western Macedonia

Integrated Master's in Electrical and Computer Engineering

Kozani, Greece

Sept 2019 – July 2024

- Top-ranked graduate of the Class of 2024.
- Awarded the "Excellence Award" by the State Scholarships Foundation (IKY) and a merit scholarship for highest academic performance (Winter 2022–2023).

PROFESSIONAL EXPERIENCE

Pfizer

Assistant Statistical Data Scientist

Thessaloniki, Greece

January 2026 – Present

- Prepared and cleaned clinical datasets by resolving duplicates and filtering irrelevant records, ensuring high data quality prior to the anonymization process.
- Utilized Blur software to execute data anonymization, masking sensitive patient variables by applying specific filters to ensure regulatory compliance.
- Managed the secure loading and migration of finalized clinical datasets into highly regulated, confidential platforms in response to internal data requests.

Infoscope Hellas

Embedded Systems Software Developer (Intern)

Thessaloniki, Greece

July 2023 – October 2023

- Developed device-agnostic C++ software to automatically decrypt and decode Wm-Bus/M-Bus sensor packets, extracting value types and measurements for real-time transmission to the company's web portal.
- Designed a multi-threaded TCP connection manager for efficient real-time sensor data transmission to the company's web portal from multiple devices.

Restaurant

Head Waiter & Waiter (Seasonal)

Chalkidiki, Greece

June 2018 – September 2025

- Managed high-volume, high-stress front-of-house operations for 8 consecutive summer seasons.
- Demonstrated exceptional adaptability, conflict resolution, and cross-team communication in a fast-paced, customer-facing environment.

PROJECTS

Smart Home Energy Data Platform (IoT + DE Pipeline) [GitHub](#)

- Built a full IoT data platform using MQTT, MongoDB (bronze), Postgres (silver/gold), and Airflow orchestrated ETLs for daily kWh/cost analytics.
- Designed a multi-service Docker architecture (simulator, ingestion, ETL, dashboard) with JSON-schema validation, incremental offsets, and real-time Streamlit dashboards.

Smart Nutrition & Wellness Data Platform (S3 + ClickHouse) — Ongoing [GitHub](#)

- Built an automated multimodal ingestion pipeline (nutrition CSV, smartwatch JSON, weather API) using MinIO (bronze), silver normalization, ClickHouse analytics, and a watchfiles-based event trigger.
- Delivered gold-level daily/weekly KPIs, rolling features, and health insights with two Streamlit dashboards (daily trends + weekly reports).

Evaluating Dimensionality Reduction Methods for ML Classifiers (Thesis) [GitHub](#)

- Developed a Python package implementing 15 DR/FS methods with automatic dimensionality estimation.

- Built a modular, research-oriented architecture enabling integration into ML pipelines and selection workflows.

Modular IoT Real-Time Data Pipeline (MQTT + HTTP ETL) [GitHub](#)

- Built a streaming and batch ETL pipeline supporting MQTT ingestion, buffered micro-batching, and scheduled HTTP data fetching.
- Implemented schema validation, device-aware enrichment, computed fields (Jinja2), and dead-letter queues.

CERTIFICATIONS

Cloud DevOps using Microsoft Azure — Udacity 2026

- Engineered scalable cloud environments utilizing Infrastructure as Code (IaC) and implemented automated CI/CD deployment pipelines via Azure DevOps.

Data Engineering with Microsoft Azure — Udacity 2026

- Architected Lakehouse solutions and automated ELT pipelines utilizing Azure Synapse, Databricks, Data Factory (ADF), and Apache Spark.

Mastering Azure Databricks for Data Engineering — Packt / Coursera 2026

- Implemented Delta Live Tables, Delta Lake optimizations (time travel), and CI/CD automated workflows via Databricks Repos and APIs.

Data Engineering Professional Certificate — DeepLearning.AI / Coursera 2025

- Deployed end-to-end data lakehouse systems and streaming pipelines on AWS leveraging Airflow, Apache Spark, and Amazon Kinesis.

IBM Data Engineering Professional Certificate — IBM / Coursera 2025

- Built robust ETL architectures and data platforms utilizing Python, Kafka, Hadoop, and Airflow across relational and NoSQL databases.

RESEARCH & PUBLICATIONS

Machine Learning Data-Based Approaches for Autism Spectrum Disorder Classification Utilizing Facial Images

Published in the 6th International Conference on ICT Integration in Technical Education (Jan 2024)

- Developed and evaluated ML-based facial recognition models for autism diagnosis, leveraging ensemble learning techniques to improve classification accuracy.

Comparison of Dimensionality Reduction and Feature Selection Methods: Enhancing Classifier Performance (Upcoming Publication, Expected 2026)

- Investigated the impact of various dimensionality reduction and feature selection techniques on classifier performance, comparing methods such as PCA, LDA, SVD, LLE, ISOMAP, T-SNE, and BORUTA.

LANGUAGES

Greek (Native); **English** (Bilingual);